Day 9 Task — Safe Malware Analysis

Time needed	Deadline	Where to submit
45 minutes	Before the Day 10 session	Portal → Day 9 → Upload

No real malware is used in this task. You will work with the EICAR test file — a harmless, industry-standard string that every antivirus is designed to flag — and with file hashes. You will never download or run actual malware.

You can do this on your host machine or your VM. If your antivirus deletes the EICAR file, that is the point — note it and move on.

Part A — Create and hash the EICAR test file

Open a plain text editor (Notepad / gedit / nano) and paste EXACTLY this one line, then save it as eicar.com (or eicar.txt if your editor won't allow .com):

```
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
```

Now compute its hash. Windows PowerShell: **Get-FileHash eicar.com -Algorithm SHA256** · Linux/macOS: **sha256sum eicar.com**

Question	Your answer
SHA-256 of your eicar file	
Did your antivirus react? What did it say/do?	
Why does AV flag a harmless file? (1 line)	

Part B — Look it up on VirusTotal

Go to **virustotal.com** → Search tab → paste the SHA-256 from Part A (search the HASH, do not upload the file). Read the Detection and Details tabs.

Question	Your answer
How many engines detect it? (e.g. 60/70)	
What name do engines give it? (e.g. EICAR-Test-File)	
What file type does VirusTotal report?	
Under ‘Details’, does the SHA-256 match yours exactly?	

Part C — Investigate an unknown hash

A colleague forwards you this SHA-256 from an alert and asks ‘are we infected?’. Search it on VirusTotal (hash only).

```
7bdd5c1c3301e9c70f9a8a9c3f6f3b7e2e6b2a1f0c9d8e7a6b5c4d3e2f10a9b8c
```

Question	Your answer
Is the hash known to VirusTotal? (found / not found)	
If found: how many detections and what family? If not found: what does 'not found' tell you?	
Would you upload the actual file to find out? Why or why not?	

Part D — Read a sandbox report

Open **any.run** public submissions (app.any.run/submissions) or **Hybrid Analysis**, open any recent public report, and skim the behaviour. You are READING a report, not running anything.

Question	Your answer
What file/sample did you look at? (name or hash)	
Name one file it created or dropped	
Name one domain or IP it contacted (its C2?)	
One persistence or registry change you noticed	

Part E — Map an attack to the kill chain

Pick ONE real attack (WannaCry, Colonial Pipeline, or the sample from Part D) and fill the chain in your own words.

Stage	What happened in this attack
Recon	
Deliver	
Exploit	
Install / Persistence	
C2 / Control	
Act (impact)	

How to submit

- One Google Doc or PDF with your four tables, the kill-chain table, and screenshots of the VirusTotal results
- Do NOT attach the eicar file to your submission (your AV may block the upload) — a screenshot is enough
- Upload under Day 9 on the portal before the Day 10 session

Marks (out of 10)

Part	Marks
A — EICAR created and hashed, AV reaction noted	2
B — VirusTotal result read correctly	2
C — unknown hash reasoning (incl. upload decision)	2
D — sandbox report behaviour noted	2
E — kill chain mapped sensibly	2

Counts toward your lab journal (30%). The upload-vs-hash reasoning in Part C matters as much as the answer. Tomorrow: OSINT — how attackers research YOU before any of this.